kassner-ODS.pdf
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Please digest these

I've digested both documents. Here is a concise technical synthesis, emphasizing the concepts most relevant to your interests in particle impacts, cold spray, dispersion strengthening, and high-temperature structural materials.

Paper 1: Creep Behavior of Particle-Strengthened Alloys (ODS and precipitate-strengthened alloys)

Source: Chapter 8, *Fundamentals of Creep in Metals and Alloys* kassner-ODS

Central Question

Why do oxide dispersion strengthened (ODS) alloys and particle-strengthened alloys retain strength during creep far beyond what classical Orowan strengthening predicts?

The chapter focuses on:

- ODS alloys (incoherent particles)
- Low-volume-fraction precipitate-strengthened alloys (<10–30%)
- Threshold stress behavior during creep

Key Concept 1: Orowan Strengthening is Not Enough

At low temperatures, strengthening is governed by:

$$\tau_{or} \approx \frac{Gb}{L}$$

where:

- G = shear modulus
- b = Burgers vector
- L = particle spacing

Particles force dislocations to bow between obstacles and leave Orowan loops. kassner-ODS

However, at elevated temperatures dislocations can climb around particles, meaning Orowan bowing alone cannot explain the creep resistance.

Key Concept 2: Threshold Stress

ODS alloys exhibit a stress below which creep becomes extremely slow:

$$\sigma_{eff} = \sigma - \sigma_{th}$$

This threshold stress is observed experimentally in Ni–Cr–ThO_2 and many ODS systems. kassner-ODS

The creep law becomes:

$$\dot{\varepsilon} = A(\sigma - \sigma_{th})^n$$

This is one of the defining features of dispersion-strengthened alloys. kassner-ODS

Key Concept 3: Two Main Explanations for Threshold Stress

A. Dislocation Climb Resistance

Dislocations climb around particles.

Climbing increases dislocation line length:

$$dE \approx \frac{Gb^2}{2} dL - \tau bLdx$$

The added line energy creates an apparent threshold stress. kassner-ODS

Two limiting cases:

Local Climb
Climb restricted near particle interfaces.

Predicted threshold:

$$\sigma_c \approx 0.19 \frac{Gb}{L}$$

General Climb
Climb extends into matrix.

Predicted threshold:

$$\sigma_c \approx 0.004 - 0.02 \frac{Gb}{L}$$

Much smaller. kassner-ODS

B. Dislocation Detachment Model

TEM observations show dislocations become trapped at particle interfaces.

After climbing around a particle, the dislocation must detach from the interface. kassner-ODS

Detachment stress:

$$\tau_d = \sqrt{1 - k_R^2} \left(\frac{Gb}{L} \right)$$

where k_R describes elastic energy relaxation at the particle interface. kassner-ODS

This predicts:

$$\tau_d \approx \frac{Gb}{3L}$$

which is substantially larger than general-climb predictions.

Today, detachment-controlled creep is considered one of the most important explanations for ODS creep resistance.

Key Concept 4: Thermally Activated Detachment

Rosler and Arzt developed a kinetic model where detachment is thermally activated. kassner-ODS

Important consequence:

The threshold stress is not truly absolute.

Instead:

- creep can still occur below the apparent threshold
- rate becomes exponentially small

This better matches experiments.

Key Concept 5: Optimum Particle Size

The detachment model predicts an optimum particle size:

Too small:

- detachment easier

Too large:

- Orowan stress decreases

Thus maximum creep resistance occurs at an intermediate particle size. kassner-ODS

This is extremely relevant for:

- ODS superalloys
- ODS steels
- advanced space nuclear materials

Key Concept 6: Coherent vs Incoherent Particles

Incoherent particles

Examples:

- Y_2O_3 dispersoids
- ThO_2 dispersoids

Mechanism:

- climb
- detachment
- interfacial pinning

Coherent particles

Examples:

- γ' in superalloys
- Al_3Sc

Can initially strengthen by:

- particle cutting
- APB formation
- coherency stresses

At elevated temperatures many eventually become climb-controlled. kassner-ODS

Most Important Takeaway

For high-temperature applications:

Particle spacing and interface characteristics matter more than simple Orowan strengthening.

The dominant creep resistance mechanism often becomes:

$$\text{climb} + \text{interfacial detachment}$$

rather than pure bowing.

This is directly relevant to:

- ODS alloys
- cold spray deposits containing oxide dispersoids
- refractory alloy design
- hypersonic materials

Paper 2: Effects of Precipitate Shape and Orientation on Dispersion Strengthening in Mg Alloys

Nie (2003) 1-s2.0-S1359646202004979-main

Central Question

For the same:

- precipitate volume fraction
- precipitate number density

which precipitate shape/orientation gives the greatest strengthening?

Key Result

Not all precipitates are equally effective.

The strengthening effectiveness depends strongly on:

1. Shape
2. Aspect ratio
3. Crystallographic orientation

not simply volume fraction. 1-s2.0-S1359646202004979-main

Orowan Framework

Strengthening:

$$\Delta\tau = \frac{Gb}{2\pi\lambda}\ln\left(\frac{d_p}{r_0}\right)$$

where λ is obstacle spacing. 1-s2.0-S1359646202004979-main

Reducing λ is the key to strengthening.

Comparison of Precipitate Types

The paper evaluates:

1. Spherical particles

Reference case.

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2. Basal plates

Plates lying parallel to the Mg basal plane.

Result:

Weak strengthening.

Increasing aspect ratio actually reduces effectiveness because the number of obstacles intersecting the slip plane decreases. 1-s2.0-S1359646202004979-main

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3. [0001] Rods

Moderately effective.

Better than spheres.

Better than basal plates. 1-s2.0-S1359646202004979-main

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4. Prismatic Plates

Most effective by far.

These plates intersect the basal slip plane nearly perpendicularly.

Dislocations encounter many more barriers. 1-s2.0-S1359646202004979-main

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Strengthening Ranking

For identical volume fraction and particle density:

$$\text{Prismatic plates} > [0001]\ \text{rods} > \text{spheres} > \text{basal plates}$$

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Aspect Ratio Effects

Prismatic plates

Strength increases dramatically with aspect ratio.

For large aspect ratios:

$$\lambda \to 0$$

meaning the effective spacing between obstacles approaches zero. 1-s2.0-S1359646202004979-main

The model predicts theoretically infinite Orowan strengthening if plates become continuous barriers.

Of course, in reality:

- local stresses rise
- precipitates shear
- other mechanisms activate

before infinity is reached.

Experimental Connection

The paper explains why:

AZ91 (Mg-Al-Zn)

Strengthened mainly by basal $Mg_{17}Al_{12}$ plates

→ relatively low strength (~120 MPa). 1-s2.0-S1359646202004979-main

WE54 (Mg-Y-RE)

Strengthened by prismatic β_1/β plates

→ much higher strength (~200 MPa). 1-s2.0-S1359646202004979-main

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Combined Insights Relevant to Your Research

Several ideas connect strongly to your work on cold spray, impact bonding, and extreme-environment alloys:

1. Interface-Controlled Strengthening

The ODS chapter shows that particle interfaces themselves can dominate high-temperature deformation through dislocation trapping and detachment.

This may be highly relevant for:

- oxide films in cold spray
- oxide fragments at particle-particle interfaces
- impact-generated nanostructures

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2. Geometry Matters as Much as Volume Fraction

The Mg paper demonstrates that orientation and aspect ratio can dominate strengthening.

For cold spray:

- elongated oxides
- lamellar oxide remnants
- impact-generated intermetallic layers

may provide much different strengthening than equivalent spherical particles.

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3. High-Temperature Design

For aerospace and hypersonics:

The ideal strengthening architecture is not necessarily:

- maximum particle fraction

but rather:

- optimized spacing
- optimized aspect ratio
- strong particle/matrix interfaces
- resistance to dislocation detachment

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4. Potential Relevance to Regolith and Space Manufacturing

The detachment/interfacial-pinning framework may provide a useful conceptual model for:

- oxide-rich extraterrestrial alloys,
- lunar regolith-derived metals,
- in-space cold spray deposits,

where nonmetallic inclusions and interfaces could act similarly to ODS dispersoids.

In short, the first document is fundamentally about **why particles improve creep resistance at high temperature**, while the second is about **how particle shape and crystallographic orientation determine Orowan strengthening efficiency**. Together they provide a useful framework linking particle geometry, interface physics, and elevated-temperature mechanical behavior.

Sources

Thought for 22s ›

Yes — but this should be treated as a **living database**, not a one-shot table. Literature values vary strongly with processing route, extrusion/rolling direction, grain morphology, annealing state, and creep-test geometry. Below is a **starter master table** organized by major ODS alloy families and classes.

ODS alloy family / class	Representative alloys	Dispersoid / particle type	Typical particle size	Typical oxide volume fraction
Ni-based ODS superalloys	MA754, MA6000, MA6000E	Y_2O_3 / Y-Al-O particles; γ′ also in MA6000	~10–50 nm	~0.5–1 vol.%
FeCrAl ODS commercial alloys	MA956, PM2000, ODM alloys	Y-Al-O, Y_2O_3-derived oxides	MA956 finest particles ~9 nm; broader 5–300 nm classes	~0.3–0.7 wt.% Y_2O_3, roughly ~0.5–1 vol.%
Ferritic 14Cr ODS / NFA steels	14YWT, 12YWT, OFRAC	Y-Ti-O nanoclusters	~2–5 nm nanoclusters, sometimes larger oxides	~0.2–0.5 wt.% Y_2O_3-equivalent
Commercial ferritic ODS steel	MA957	Y-Ti-O nanoclusters + $Y_2Ti_2O_7$	~2–15 nm clusters; ~15–35 nm oxides	~0.25 wt.% Y_2O_3 + Ti additions
ODS EUROFER / RAFM steels	ODS-EUROFER, 9Cr ODS	Y_2O_3 with Ti, Zr, Hf, Sc, La, Ce modifications	~5–13 nm	~0.3 wt.% Y_2O_3 typical

ODS alloy family / class	Representative alloys	Dispersoid / particle type	Typical particle size	Typical oxide volume fraction
High-volume-fraction ODS Fe-Al systems	Fe-10Al-4Y_2O_3, Fe-11Al-1O, Fe-17Cr-7Al-4Y_2O_3	High-volume-fraction Y-rich oxides	typically nanoscale, often ~5–30 nm	~4–7 vol.% oxide class
Co/Ni/Fe high-entropy or multi-principal ODS alloys	ODS HEAs, ODS CoCrFeNi variants	Y_2O_3-derived oxides, complex oxides	~5–50 nm	~0.5–2 vol.%
ODS Cu alloys	Cu-Al_2O_3, GlidCop-type, AM ODS Cu	Al_2O_3, Y_2O_3, oxide networks	~10–100 nm; some commercial dispersoids ~30–40 nm	~0.1–1.1 wt.% Al_2O_3 typical
ODS Al alloys	Al-MgO, Al-Al_2O_3, Al-Y_2O_3, AM ODS Al	MgO, Al_2O_3, Y_2O_3	Recent MgO ODS Al: ~5 nm	variable; often low vol.% but high number density
ODS Ti alloys	ODS Ti, Ti-Y_2O_3, AM Ti-ODS	Y_2O_3, Ti-O-Y particles	~5–100 nm	usually <1 vol.%
ODS γ-TiAl / titanium aluminides	Ti-45Al-3Nb-Y_2O_3, ODS γ-TiAl	Y_2O_3-derived particles; faceted oxides possible	~10–100 nm depending SPS/AM	<0.2 at.% Y_2O_3 in one reported alloy
ODS Mo alloys	Mo-La_2O_3, Mo-Y_2O_3, TZM-ODS variants	La_2O_3, Y_2O_3	~10–500 nm depending route	~0.5–1 wt.% La_2O_3 common
ODS W alloys	W-La_2O_3, W-Y_2O_3, W-TiC-Y_2O_3	La_2O_3, Y_2O_3, mixed oxides	~10–500 nm	~0.5–2 wt.% oxide
ODS Pt / Pt-Rh alloys	Pt-ZrO_2, Pt-Rh-ZrO_2	ZrO_2	Pt-Rh-ZrO_2: ~5–35 nm	often ~0.1–0.5 vol.%
ODS superalloy derivatives / Alloy 617 ODS	Alloy 617 ODS, Inconel ODS variants	Y-Al-O, complex Ni-Y-Al-O oxides	fine nanoscale oxides	~0.5–1 vol.%
AM-enabled ODS alloys	AM ODS Cu, AM ODS Ti, AM ODS steels, AM ODS superalloys	oxide particles retained or redistributed during AM	~5–100 nm	designed by powder feedstock

For Denise’s spreadsheet, I would add these columns immediately: **alloy family, exact composition, processing route, heat treatment, grain size/aspect ratio, texture/orientation, particle chemistry, particle diameter distribution, number density, volume fraction, interparticle spacing, yield strength temperature, yield strength, creep temperature, creep stress, rupture time, minimum creep rate, threshold stress, stress exponent, activation energy, source DOI, and notes on whether values are longitudinal or transverse.**

The most useful “paper-ready” normalized fields will be:

$$\sigma_y/G, \quad \sigma_{th}/G, \quad \sigma_{th}/\sigma_{or}, \quad L/b, \quad r/b, \quad f, \quad \dot{\varepsilon}_{min}, \quad t_r, \quad T/T_m$$

That will let the table feed directly into Ashby plots rather than remain just a literature summary.

Sources

I'm doing a review paper and I want to make sure I get the bifurcation of these alloy families correct. What should be the major class of ODS alloys and sub-families that I should keep in mind? So just give me this organization or bullet point list of organization bifurcation.

For a review paper that aims to become a **design framework for ODS alloys**, I would avoid organizing purely by base metal (Fe, Ni, Ti, etc.). Instead, I would use a **hierarchical classification** that starts with strengthening architecture and then branches into alloy systems.

This is the structure I would use.

Level 1: Dispersion-Strengthened Alloy Classes

Class I: Classical Oxide Dispersion Strengthened (ODS) Alloys

Mechanically alloyed alloys containing intentionally introduced stable oxide particles.

Examples:

- MA754
- MA956
- PM2000
- MA957
- 14YWT
- ODS EUROFER

Characteristics:

- Y_2O_3-derived dispersoids
- Typically 2–50 nm particles
- Threshold-stress creep behavior
- Climb/detachment-controlled creep

Class II: Nano-Feature Alloys (NFA)

Subset of advanced ODS alloys containing ultra-fine nanoclusters.

Examples:

- 12YWT
- 14YWT
- OFRAC alloys

Characteristics:

- Y-Ti-O nanoclusters
- 1–5 nm features
- Extremely high sink strength
- Radiation resistant

Applications:

- Fusion
- Advanced fission reactors

Class III: Internally Oxidized Alloys

Oxides formed in situ rather than introduced by mechanical alloying.

Examples:

- Cu-Al_2O_3
- Ag-CdO
- Some Fe-Al-O systems

Characteristics:

- Thermodynamically generated dispersions

- Different particle/matrix interfaces from classical ODS

Class IV: Oxide-Stabilized High Entropy and Complex Concentrated Alloys

Emerging class.

Examples:

- ODS CoCrFeMnNi
- ODS refractory HEAs

Characteristics:

- Complex matrix chemistry
- Nanoscale oxide dispersions
- Multiple strengthening mechanisms

Level 2: Base Alloy Family

Within each class:

A. Fe-Based ODS Alloys

A1. Ferritic ODS Steels

Examples:

- MA956
- PM2000

Applications:

- Turbines
- Heat exchangers

A2. Ferritic-Martensitic ODS Steels

Examples:

- MA957
- 9Cr ODS
- EUROFER-ODS

Applications:

- Nuclear systems

A3. Nanostructured Ferritic Alloys (NFAs)

Examples:

- 12YWT
- 14YWT

Applications:

- Fusion reactors

A4. FeCrAl ODS Alloys

Examples:

- PM2000

- MA956

Applications:

- Accident tolerant fuels
- Oxidation-resistant systems

B. Ni-Based ODS Alloys

B1. γ Matrix ODS Alloys

Examples:

- MA754

Characteristics:

- Pure ODS strengthening

B2. γ + γ′ ODS Superalloys

Examples:

- MA6000
- MA6000E

Characteristics:

- ODS + precipitation strengthening

These are particularly important because they combine:

- γ′ strengthening
- oxide strengthening
- grain morphology control

C. Co-Based ODS Alloys

Examples:

- ODS CoCr alloys

Less common but increasingly important.

D. Ti-Based ODS Alloys

D1. α/β Titanium ODS

Examples:

- Ti-Y_2O_3

Applications:

- Aerospace

D2. γ-TiAl ODS

Examples:

- TiAl-Y_2O_3 systems

Important for your review because:

These often contain:

- faceted particles
- anisotropic particles
- morphology-controlled creep

This is where the Nie paper becomes very relevant.

E. Refractory ODS Alloys

E1. Mo-Based ODS

Examples:

- Mo-La_2O_3
- Mo-Y_2O_3

E2. W-Based ODS

Examples:

- W-Y_2O_3
- W-La_2O_3

Applications:

- Fusion plasma-facing materials

E3. Nb-Based ODS

Examples:

- Nb-Y_2O_3

Still emerging.

F. Precious Metal ODS Alloys

Examples:

- Pt-ZrO_2
- Pt-Rh-ZrO_2

Historically important because many threshold-stress theories were tested in these systems.

G. Lightweight ODS Alloys

G1. Al-Based ODS

Examples:

- SAP alloys
- Al-Al_2O_3
- Al-MgO

G2. Mg-Based ODS

Rare but growing.

Interesting because particle orientation effects become important.

Level 3: Particle Morphology Classification

This is where I think your review can become unique.

Instead of treating all ODS alloys as spherical-particle systems, classify them by obstacle geometry.

Type I: Spherical Dispersoids

Examples:

- Y_2O_3 in MA754
- Y-Ti-O clusters in 14YWT

Characterized by:

$$AR \approx 1$$

Type II: Faceted Dispersoids

Examples:

- $Y_2Ti_2O_7$
- complex pyrochlores
- γ-TiAl oxides

Characterized by:

- strong anisotropic interfaces
- orientation-dependent detachment

Type III: Rod-Like Dispersoids

Examples:

- elongated oxides after extrusion

Characterized by:

$$AR > 5$$

Type IV: Plate-Like Dispersoids

Examples:

- layered oxides
- some oxide precipitates

Most directly connected to Nie's shape-factor framework.

Type V: Cluster Strengthened Systems

Examples:

- 14YWT
- 12YWT

Obstacle dimensions:

$$1 - 5 \text{ nm}$$

These may behave differently from classical ODS particles.

Level 4: Strengthening Mechanism Classification

This may actually become the most important figure in the review.

Regime A: Orowan-Controlled

Low temperature.

$$\tau \sim Gb/L$$

Regime B: Climb-Controlled

Intermediate temperature.

Regime C: Detachment-Controlled

Classical ODS creep regime.

$$\tau_d \sim \frac{Gb}{L}$$

modified by relaxation factor k_R.

Regime D: Shape-Controlled

Faceted particles
Plate particles
Rod particles

(Nie-inspired regime)

Regime E: Interface-Controlled

Particle chemistry dominates:

- Y-Ti-O
- Y-Al-O
- Y-Hf-O
- Y-Zr-O

rather than particle size alone.

The final taxonomy I would use in the review

ODS Alloys

1. Fe-based
   - Ferritic
   - Ferritic-Martensitic
   - NFA
   - FeCrAl
2. Ni-based
   - γ ODS
   - γ + γ′ ODS
3. Co-based
4. Ti-based
   - α/β Ti ODS
   - γ-TiAl ODS
5. Refractory

- Mo
- W
- Nb

6. Lightweight
   - Al
   - Mg
7. Precious-metal ODS
   - Pt
   - Pt-Rh

Then for every alloy in the database, add tags:

- Particle chemistry
- Particle size
- Volume fraction
- Shape (sphere/rod/plate/faceted)
- Interface type
- Threshold stress
- Relaxation factor (if available)
- Creep mechanism

That organization will support essentially every Ashby map we discussed and will naturally lead to a "materials genome" style design framework for ODS alloys rather than just another alloy-by-alloy review.

a quick table version

Level 1 Class	Level 2 Family	Representative Alloys	Primary Oxide/Dispersoid	Primary Application
Classical ODS Alloys	Ferritic ODS Steels	MA956, PM2000	Y-Al-O, Y_2O_3	High-temperature oxidation-resistant components
	Ferritic-Martensitic ODS Steels	MA957, ODS EUROFER, 9Cr ODS	Y-Ti-O, Y_2O_3	Nuclear reactors
	Nanostructured Ferritic Alloys (NFA)	12YWT, 14YWT	Y-Ti-O nanoclusters	Fusion & advanced fission
	FeCrAl ODS Alloys	PM2000, MA956	Y-Al-O	Accident-tolerant fuels, oxidation resistance
	Ni-based γ ODS Alloys	MA754	Y_2O_3	Gas turbines, creep-resistant components
	Ni-based γ + γ′ ODS Superalloys	MA6000, MA6000E	Y_2O_3 + γ′ precipitates	Advanced turbine blades
	Co-based ODS Alloys	ODS CoCr alloys	Y_2O_3	Emerging high-temperature alloys
	Ti-based ODS Alloys	Ti-Y_2O_3	Y_2O_3	Aerospace
	ODS γ-TiAl Alloys	TiAl-Y_2O_3 systems	Y_2O_3, faceted oxides	Aerospace engines
	Mo-based ODS Alloys	Mo-La_2O_3, Mo-Y_2O_3	La_2O_3, Y_2O_3	Ultra-high-temperature applications
	W-based ODS Alloys	W-Y_2O_3, W-La_2O_3	Y_2O_3, La_2O_3	Fusion plasma-facing materials

Level 1 Class	Level 2 Family	Representative Alloys	Primary Oxide/Dispersoid	Primary Application
	Nb-based ODS Alloys	Nb-Y_2O_3	Y_2O_3	Emerging refractory systems
	Al-based ODS Alloys	SAP, Al-Al_2O_3	Al_2O_3	Lightweight high-temperature alloys
	Mg-based ODS Alloys	Mg-Y_2O_3 systems	Y_2O_3	Emerging lightweight systems
	Precious Metal ODS Alloys	Pt-ZrO_2, Pt-Rh-ZrO_2	ZrO_2	Glass processing, extreme-temperature environments
Internally Oxidized Alloys	Cu-based Internal Oxidation	Cu-Al_2O_3, GlidCop	Al_2O_3	High conductivity + strength
	Ag-based Internal Oxidation	Ag-CdO, Ag-SnO_2	CdO, SnO_2	Electrical contacts
ODS High-Entropy / CCA Alloys	ODS HEAs	ODS CoCrFeMnNi	Y_2O_3	Emerging structural materials
	ODS Refractory HEAs	ODS NbMoTaW systems	Y_2O_3	Ultra-high-temperature applications
Additive-Manufactured ODS Alloys	AM ODS Steels	AM 9Cr ODS, AM EUROFER	Y-Ti-O	Advanced nuclear systems
	AM ODS Superalloys	AM ODS Inconel variants	Y-Al-O	Turbines
	AM ODS Ti Alloys	AM Ti-Y_2O_3	Y_2O_3	Aerospace
	AM ODS Cu Alloys	AM GlidCop variants	Al_2O_3	Thermal management

Cross-Cutting Classification Tags (apply to every alloy)

Category	Subclasses
Particle Morphology	Spherical, Faceted, Rod-like, Plate-like, Nanoclusters
Particle Chemistry	Y_2O_3, Y-Ti-O, Y-Al-O, Y-Hf-O, Y-Zr-O, La_2O_3, ZrO_2, Al_2O_3
Strengthening Regime	Orowan-controlled, Climb-controlled, Detachment-controlled, Interface-controlled, Shape-controlled
Matrix Type	BCC, FCC, HCP, Intermetallic
Primary Design Objective	Creep, Radiation tolerance, Oxidation resistance, Specific strength, Thermal conductivity

For the review paper, I would recommend making the **main organization by matrix family**:

Fe-based → Ni-based → Co-based → Ti-based → Refractory (Mo/W/Nb) → Al/Mg lightweight → Precious metal → HEA/CCA → Additive-manufactured ODS

and then within each family classify by:

particle chemistry → particle morphology → creep mechanism.

That hierarchy is what most readers (especially from metallurgy, nuclear materials, and high-temperature alloys) will expect, while still allowing you to build the broader Ashby-style design framework.